Radial Projection — Definition and Relation to Central Projection

A Foundational Mapping in the Central Projection Framework
(Technical Note)

Noriaki Kihara

2026-05

Concept DOI (auto-redirects to the latest version): 10.5281/zenodo.20462569 **v3.2 Version DOI**: 10.5281/zenodo.20500187 **Zenodo page**: https://zenodo.org/records/20500187

Character of This Note

This note is a technical note that explicitly defines and organizes the foundational mapping underlying the central projection framework of [1], [2].

The mapping treated in this note,

$$\sigma_R : \mathbb{R}^{n+1} \setminus \{0\} \to S^n(R), \quad \sigma_R(x) = \frac{R}{\|x\|} \cdot x$$

is classically known in topology as the **radial projection**, in linear algebra and differential geometry as the **normalization map**, and as a textbook example of a deformation retract ([Hatcher, Ch. 0]). The note does not claim any novel geometric theorem; it only does the following:

- It names this mapping **"radial projection σ_R"** and fixes the notation (§2).
- It shows that the author's central projection [1] and composition operation [2] can be characterized as the restriction of the radial projection to the tangent hyperplane Π_R (§3).
- It organizes the contrast between σ_R and the central projection Φ_R with respect to injectivity, image, and differential structure (§3).

The note is not submitted to a journal; it is released as a **Zenodo preprint that consolidates the foundations** of the author's central projection series.

§1 Introduction

1.1 Motivation

The mapping $x \mapsto (R/\|x\|)x$, which normalizes a vector and sends it to the sphere of prescribed radius, is classically known in topology as the **radial projection** (or **radial retraction**) and in linear algebra as the **normalization map**. In Hatcher's *Algebraic Topology*, Chapter 0, for instance, it appears as a canonical example of a deformation retract from $\mathbb{R}^n \setminus \{0\}$ onto S^{n-1} and is treated as one of the most basic examples of a homotopy equivalence.

In this note we fix the symbol σ_R for this already-known mapping, the symbol used within the author's central projection framework [1], [2], and we collect the basic properties needed for that framework in one place. Since many textbooks treat this mapping only as an intermediate computation or example, an explicit note is useful as a reference base within the series.

The purpose of this note is to name this foundational mapping "radial projection σ_R", to fix its notation, and to make explicit its relation to the operations appearing in the author's central projection framework [1], [2].

1.2 Relation to the Existing Central Projection Framework

In the preceding paper [1] the author gave a geometric formulation of four-dimensional spacetime via the central projection $\Phi_R : \Pi_R \to S^n(R)$. In [2] it was further shown that dimension reduction operations along several axes are correctly algebraized not as "multiple central projections" but as **a single central projection together with a commuting section on the sphere**.

This note shows that these existing central projections can be characterized as **the restriction to the tangent hyperplane** Π_R of the radial projection σ_R, which has a larger domain (Lemma 3.2).

A remark on orientation: the "central projection" $\Phi_R : \Pi_R \to S^n(R)$ of [1] has the **opposite orientation** to the gnomonic projection of classical cartography ($S^n_+ \to \Pi_R$). In this note and in [1] we adopt the convention that the tangent hyperplane is the source and the sphere is the target.

1.3 Scope of This Note

This note treats only the following:

- The definition and basic properties of the radial projection σ_R.
- Its relation to the central projection Φ_R (its rôle as a specialization).
- The contrast between the image and injectivity of Φ_R and the non-injectivity of σ_R.

The following lie outside the scope of this note:

- Physical interpretations or applications of the radial projection (curvature, spacetime, quantum theory, etc.).
- Concrete applications of [1], [2] (induced metric, Einstein tensor, black-hole thermodynamics, etc.).
- Correspondences with abstract submanifold theory such as isoparametric functions or polar actions.
- Comparisons with other sphere-related mappings such as stereographic projection (which is conformal).

These will be addressed in separate notes or separate papers.

1.4 A Note on Terminology

The term "spherical projection" is used in the literature with various meanings (stereographic projection, various map projections, etc.). In this note we refer to σ_R by the name **radial projection**, which is the standard term in topology. To avoid confusion, we mention this terminological choice up front.

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§2 Definition and Basic Properties of the Radial Projection

2.1 Definition

Definition 2.1 (Radial projection). For a positive real number $R > 0$, the radial projection σ_R is defined by

$$\sigma_R : \mathbb{R}^{n+1} \setminus \{0\} \to S^n(R), \quad \sigma_R(x) = \frac{R}{\|x\|} \cdot x \tag{2.1}$$

where $S^n(R) = \{y \in \mathbb{R}^{n+1} \mid \|y\| = R\}$ is the n-dimensional sphere of radius R, and $\|\cdot\|$ denotes the standard Euclidean norm.

Well-definedness. Since $\|\sigma_R(x)\| = (R/\|x\|) \cdot \|x\| = R$, we have $\sigma_R(x) \in S^n(R)$.

2.2 Basic Properties

Proposition 2.1 (Basic properties).

(i) σ_R is of class C^∞ on the whole of $\mathbb{R}^{n+1} \setminus \{0\}$ (as a smooth map into the submanifold $S^n(R) \subset \mathbb{R}^{n+1}$).

(ii) σ_R is a **retraction onto** $S^n(R)$: $\sigma_R\big|_{S^n(R)} = \mathrm{id}_{S^n(R)}$. In particular, it is surjective.

(iii) For any $t > 0$ and any $x \in \mathbb{R}^{n+1} \setminus \{0\}$, $\sigma_R(tx) = \sigma_R(x)$.

Proof.

(i) Since $\|x\| > 0$, the map $1/\|x\|$ is C^∞, the linear map $x \mapsto x$ is C^∞, and their product is C^∞.

(ii) For $y \in S^n(R)$ we have $\|y\| = R$, hence $\sigma_R(y) = (R/R)y = y$. Therefore $S^n(R) \subset \mathrm{Im}(\sigma_R)$, i.e. σ_R is surjective.

(iii) $\sigma_R(tx) = (R/\|tx\|)(tx) = (R/(t\|x\|))(tx) = (R/\|x\|)x = \sigma_R(x)$. $\square$

Footnote (the case $t < 0$). For $t < 0$ one has $\sigma_R(tx) = -\sigma_R(x)$, so the orientation is reversed. Property (iii) identifies only the **positive ray** $\{tx : t > 0\}$ emanating from the origin, not the whole line through the origin.

2.3 Idempotency and Retract Structure

Proposition 2.2 (Idempotency). Regarding σ_R as a self-map of $\mathbb{R}^{n+1} \setminus \{0\}$, one has $\sigma_R \circ \sigma_R = \sigma_R$.

Proof. Since $\sigma_R(x) \in S^n(R)$, Proposition 2.1(ii) gives $\sigma_R(\sigma_R(x)) = \sigma_R(x)$. $\square$

This justifies the term “projection”: σ_R is a projection operator onto its image $S^n(R)$.

Deformation retract. The homotopy $H(x, s) = (1 - s)x + s \cdot \sigma_R(x)$ ($s \in [0, 1]$) realizes a **strong deformation retract** of $\mathbb{R}^{n+1} \setminus \{0\}$ onto $S^n(R)$ (classical; see [Hatcher, Ch. 0]).

2.4 Characterization as a Quotient Space

The multiplicative group $\mathbb{R}_{>0}$ of positive reals acts on $\mathbb{R}^{n+1} \setminus \{0\}$ by scalar multiplication $t \cdot x = tx$.

Proposition 2.3 (Quotient space). The quotient space

$$(\mathbb{R}^{n+1} \setminus \{0\})/\mathbb{R}_{>0} \cong S^n(R)$$

and σ_R may be interpreted as the canonical section that selects, from each positive ray (orbit), the representative point of norm R.

Proof. The quotient map $\pi : \mathbb{R}^{n+1} \setminus \{0\} \to (\mathbb{R}^{n+1} \setminus \{0\})/\mathbb{R}_{>0}$ and σ_R have the same fibers $\{tx : t > 0\}$, so an induced map $\overline{\sigma_R} : (\mathbb{R}^{n+1} \setminus \{0\})/\mathbb{R}_{>0} \to S^n(R)$ is well-defined and is a bijection. Continuity and smoothness are standard. $\square$

This is a structural restatement of Proposition 2.1(iii); the quotient structure makes explicit that **the radial information is completely discarded**.

2.5 Differential Structure

Proposition 2.4 (Kernel and image of the differential). At each $x \in \mathbb{R}^{n+1} \setminus \{0\}$, the differential $D\sigma_R|_x : \mathbb{R}^{n+1} \to T_{\sigma_R(x)}S^n(R)$ of σ_R is

$$D\sigma_R|_x(v) = \frac{R}{\|x\|}\left(v - \frac{x \cdot v}{\|x\|^2}x\right) \tag{2.2}$$

and its kernel is

$$\ker D\sigma_R|_x = \operatorname{span}\{x\} \tag{2.3}$$

namely the x-direction (the radial direction). The map $D\sigma_R|_x$ is a submersion of rank n, and its image is

$$\operatorname{Im}(D\sigma_R|_x) = x^\perp = T_{\sigma_R(x)}S^n(R) \tag{2.4}$$

(since $\sigma_R(x)$ is a positive scalar multiple of x, the tangent space of $S^n(R)$ at $\sigma_R(x)$ equals the orthogonal complement $x^\perp$ of x).

Proof. Using that the directional derivative of $\|x\|^{-1}$ along v equals $-\|x\|^{-3}(x \cdot v)$, differentiating $\sigma_R(x) = R\|x\|^{-1}x$ along v gives

$$D\sigma_R|_x(v) = R\big(-\|x\|^{-3}(x \cdot v)\big)x + R\|x\|^{-1}v = -R\|x\|^{-3}(x \cdot v)x + R\|x\|^{-1}v,$$

which rearranges to (2.2). For the kernel, setting $D\sigma_R|_x(v) = 0$ yields

$$v - \frac{x \cdot v}{\|x\|^2}x = 0,$$

hence

$$v = \frac{x \cdot v}{\|x\|^2}x,$$

so v is a scalar multiple of x. Conversely, if $v = \lambda x$, then

$$v - \frac{x \cdot v}{\|x\|^2}x = \lambda x - \frac{\lambda\|x\|^2}{\|x\|^2}x = 0,$$

so $D\sigma_R|_x(v) = 0$. Therefore $\ker D\sigma_R|_x = \operatorname{span}\{x\}$.

For the image, for any v we have $x \cdot D\sigma_R|_x(v) = \frac{R}{\|x\|}\left(x \cdot v - \frac{x \cdot v}{\|x\|^2}\|x\|^2\right) = 0$, so $\mathrm{Im}(D\sigma_R|_x) \subseteq x^\perp$. By the rank–nullity theorem, $\dim \mathrm{Im}(D\sigma_R|_x) = (n+1) - \dim\ker D\sigma_R|_x = n = \dim x^\perp$, hence $\mathrm{Im}(D\sigma_R|_x) = x^\perp = T_{\sigma_R(x)}S^n(R)$. $\square$

This clarifies the differential-geometric structure of the radial projection: **the radial direction corresponds exactly to the kernel of the differential, and only the tangential directions are mapped onto the sphere**.

2.6 Preservation of Angles Between Vectors

Proposition 2.5 (Preservation of angles). For any $x, y \in \mathbb{R}^{n+1} \setminus \{0\}$,

$$\angle(\sigma_R(x), \sigma_R(y)) = \angle(x, y) \tag{2.5}$$

where $\angle(\cdot,\cdot)$ denotes the unoriented angle between vectors.

Proof.

$$\frac{\sigma_R(x) \cdot \sigma_R(y)}{\|\sigma_R(x)\| \cdot \|\sigma_R(y)\|} = \frac{(R/\|x\|)(R/\|y\|)(x \cdot y)}{R \cdot R} = \frac{x \cdot y}{\|x\| \cdot \|y\|},$$

so the cosines agree and the angles are equal. $\square$

Remark (terminology). This is the preservation of the angle between two vectors and is a different notion from conformality (preservation of angles between curves). Since σ_R drops dimension, preservation of angles between curves is not even meaningful for it.

2.7 Scale Invariance

Proposition 2.6 (Invariance of direction under radius rescaling). For any $R_1, R_2 > 0$ and any $x \in \mathbb{R}^{n+1} \setminus \{0\}$, the points $\sigma_{R_1}(x)$ and $\sigma_{R_2}(x)$ lie on the same positive ray from the origin O through x.

Proof. Both $\sigma_{R_1}(x) = (R_1/\|x\|)x$ and $\sigma_{R_2}(x) = (R_2/\|x\|)x$ are positive scalar multiples of x. $\square$

That is, in the radial projection **the directional component is invariant under rescaling of the radius** R.

Fig. 1: Illustration of the radial projection σ_R ($n = 1$, $R = 3$). With the origin O as the viewpoint, the points A (inside the circle) and B, C (outside the circle, in different quadrants) on radial rays are sent to the points A', B', C' on the circle.

Fig. 3: Invariance of direction under radius rescaling in the radial projection (Proposition 2.6). For both radii $R = 3$ (black circle) and $R = 5$ (purple circle), the images of A, B, C lie on the same radial rays. Consequently the

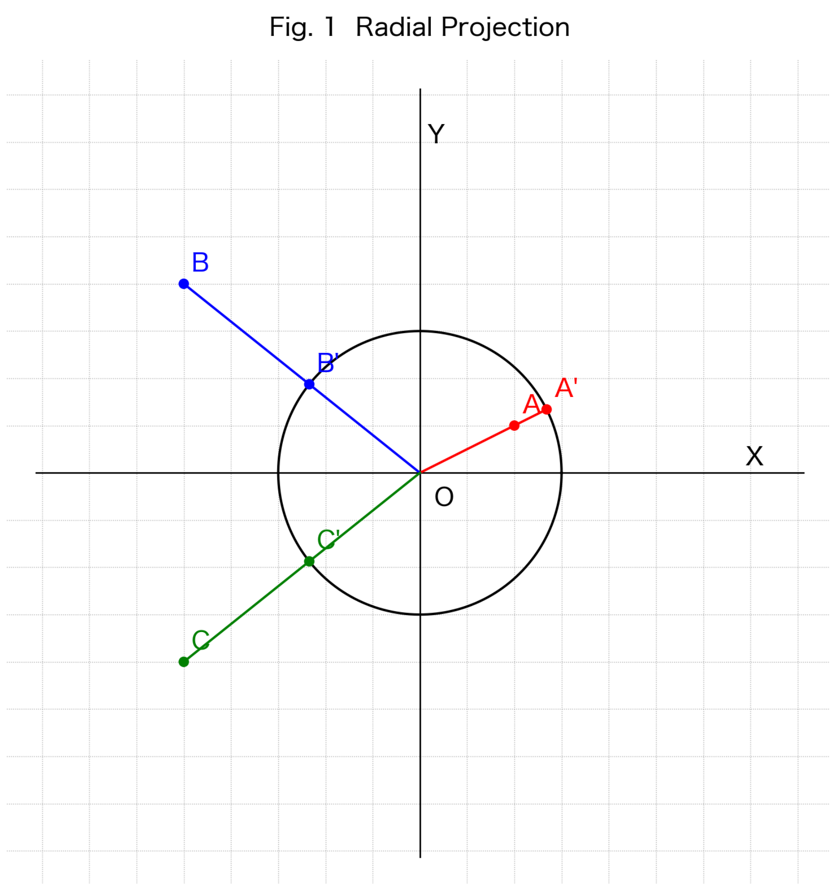

Figure 1: Fig. 1 Radial Projection

Fig. 3 Radial Projection (double R)

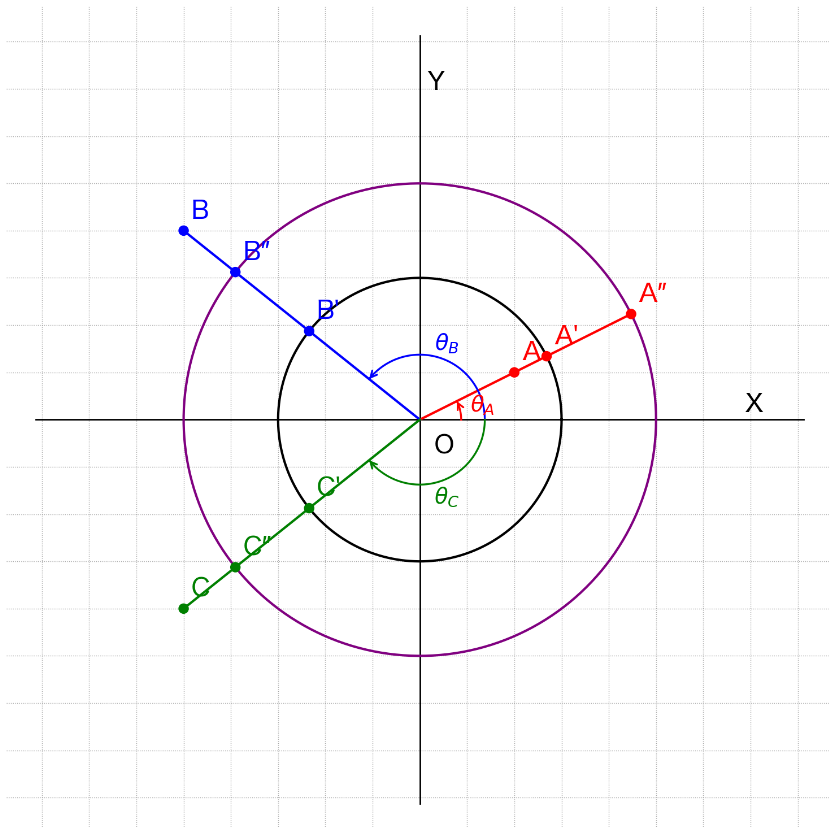

Figure 2: Fig. 3 Radial Projection (double R)

angles $\theta_A, \theta_B, \theta_C$ that each point makes with the positive x-axis are preserved independently of R.

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§3 Relation to the Central Projection

3.1 Definition of the Central Projection

Definition 3.1 (Central projection). Let the tangent hyperplane at the north pole $N = (0, \ldots, 0, R)$ of the sphere $S^n(R)$ be

$$\Pi_R = \{x \in \mathbb{R}^{n+1} \mid x_{n+1} = R\}. \tag{3.1}$$

We have $\Pi_R \cong \mathbb{R}^n$. The central projection Φ_R is defined by

$$\Phi_R : \Pi_R \to S^n_+(R), \quad \Phi_R(x) = \frac{R}{\|x\|} \cdot x \tag{3.2}$$

where $S^n_+(R) = \{y \in S^n(R) \mid y_{n+1} > 0\}$ is the **open upper hemisphere**. That this is the image of Φ_R is shown in Proposition 3.1 below.

This agrees with the central projection defined in §2 of [1] (for the positivity of the final coordinate, see Remark 3.1 of [1]).

Remark (on orientation). The present Φ_R has the orientation **tangent hyperplane (source) $\to$ sphere (target)**. The gnomonic projection of classical cartography is often defined in the opposite direction (sphere $\to$ tangent plane), so the terminological mismatch should be kept in mind.

3.2 Image and Injectivity of the Central Projection

Proposition 3.1 (Image and injectivity of Φ_R).

(i) The image of Φ_R is the open upper hemisphere $S^n_+(R) = \{y \in S^n(R) \mid y_{n+1} > 0\}$.

(ii) $\Phi_R : \Pi_R \to S^n_+(R)$ is a **diffeomorphism** (in particular, injective).

Proof.

(i) For $x \in \Pi_R$ we have $x_{n+1} = R$, so the last coordinate of $\Phi_R(x)$ is $(R/\|x\|) \cdot R = R^2/\|x\| > 0$. Hence $\Phi_R(\Pi_R) \subset S^n_+(R)$. Conversely, given any $y \in S^n_+(R)$ with $y_{n+1} > 0$, set $x := (R/y_{n+1})y$; then $x_{n+1} = R$, so $x \in \Pi_R$, $\|x\| = R\|y\|/y_{n+1} = R^2/y_{n+1}$, and $\Phi_R(x) = (R/\|x\|)x = (y_{n+1}/R) \cdot (R/y_{n+1})y = y$. Hence Φ_R is surjective onto $S^n_+(R)$.

(ii) The construction in (i) furnishes the inverse $\Phi_R^{-1}(y) = (R/y_{n+1})y$, which is C^∞ in both directions. $\square$

3.3 Relation Between the Radial Projection and the Central Projection

Lemma 3.2 (The central projection is a restriction of the radial projection). The central projection Φ_R coincides with the radial projection σ_R when the domain is restricted to Π_R and the codomain is restricted to $S^n_+(R)$:

$$\Phi_R = \sigma_R\big|_{\Pi_R} : \Pi_R \to S^n_+(R). \tag{3.3}$$

Proof. The defining formulas $\sigma_R(x) = (R/\|x\|)x$ (Definition 2.1) and $\Phi_R(x) = (R/\|x\|)x$ (Definition 3.1) are identical. Moreover $\Pi_R \subset \mathbb{R}^{n+1} \setminus \{0\}$ (since on Π_R we have $x_{n+1} = R > 0$, so no point of Π_R is the origin), and restricting the domain of σ_R to Π_R gives Φ_R. That the image lies in $S^n_+(R)$ follows from Proposition 3.1(i). □

3.4 Contrast Between σ_R and Φ_R

Property	σ_R	Φ_R
Domain	$\mathbb{R}^{n+1} \setminus \{0\}$	Π_R (tangent hyperplane, isomorphic to $\mathbb{R}^n$)
Image	$S^n(R)$ (whole sphere)	$S^n_+(R)$ (**open upper hemisphere only**)
Injectivity	**Non-injective** (each ray collapses to a point)	**Injective** (diffeomorphism)
Radial direction	Contained in the kernel (Proposition 2.4)	The kernel direction is not contained in the domain's tangent space (transversal)

This contrast is the **core** of the present note. The radial projection extends the domain to all of $\mathbb{R}^{n+1} \setminus \{0\}$ at the cost of losing injectivity. The central projection, on the other hand, retains injectivity and the rich geometric structure (induced metric, curvature, etc.) but its domain is restricted to Π_R and its image to $S^n_+(R)$.

Fig. 2: Illustration of the central projection Φ_R ($n = 1$, $R = 3$). The points A'', B'' on the tangent hyperplane (purple line; for $n = 1$ this is the line $y = R$, and in general it is $x_{n+1} = R$ as in (3.1)) tangent to the sphere at the north pole are projected onto A', B' on the sphere along radial rays from the origin O. The radial projection σ_R is defined along the very same rays, and the sole difference between the two is their domain. The image of Φ_R is the open upper hemisphere (it does not reach the lower hemisphere, nor does it reach the equator from any finite point). The green pair C, C' is an example of the radial projection σ_R and is not contained in the image of the central projection Φ_R (the open upper hemisphere).

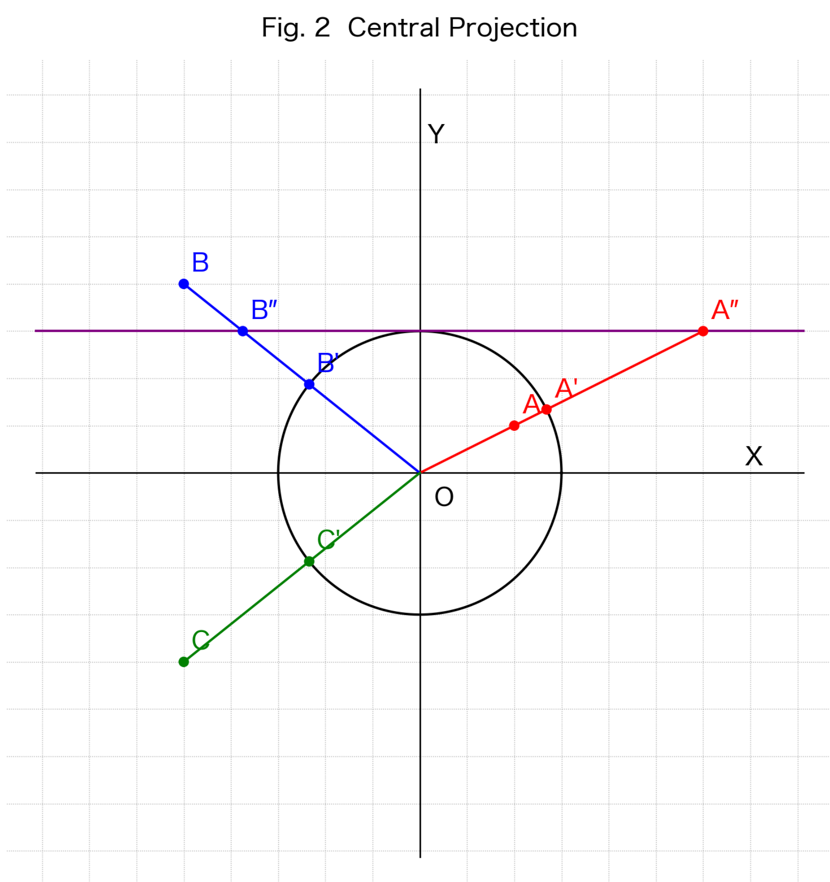

Figure 3: Fig. 2 Central Projection

3.5 Relation to Existing Results

Remark 3.1 (Rewriting of existing results). The geometric quantities defined via Φ_R in [1] and [2] (the induced metric $g_{\mu\nu} = \Phi_R^* g_{S^n(R)}$, the Einstein tensor, the composite curvature radius $r_{\text{final}}^2 = r_1^2 - \sum_{j\in S}(x_{i_j}^*)^2$, and so on) can be rewritten in the notation of this note as pullbacks under $\sigma_R\big|_{\Pi_R}$.

However, these geometric quantities are intrinsically structures on Π_R (mapping into the upper hemisphere) and do not extend automatically to the extended domain of the radial projection σ_R. For a general smooth map $f : M \to N$, the pullback f^*T is well-defined regardless of injectivity; in particular $\sigma_R^* g_{S^n(R)}$ is a well-defined symmetric 2-tensor on $\mathbb{R}^{n+1}\setminus\{0\}$. But, by Proposition 2.4, the radial direction $\text{span}\{x\}$ lies in the kernel of the differential, so this pullback tensor is degenerate and is not a Riemannian metric. Hence it cannot be identified with the non-degenerate induced metric that $\Phi_R^* g_{S^n(R)}$ gives on Π_R.

The semigroup structure, commutativity, and closed form of the composite curvature radius for the section operation of [2] (which, to avoid notational collision in this note, we denote κ_S) are positioned as operations on the image space $S^n(R)$ of σ_R.

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§4 Conclusion

4.1 Main Results of This Note

1. We have explicitly defined the radial projection $\sigma_R(x) = (R/\|x\|)x$ as a C^∞ retraction from $\mathbb{R}^{n+1} \setminus \{0\}$ onto $S^n(R)$ (Definition 2.1, Proposition 2.1).
2. We have organized at an elementary level its idempotency, deformation retract structure, quotient-space description, kernel and image of the differential, angle preservation, and scale invariance (Propositions 2.2–2.6).
3. The central projection Φ_R is equal to the restriction of the radial projection σ_R to the tangent hyperplane Π_R, its image is the **open upper hemisphere** $S_+^n(R)$, and Φ_R is a diffeomorphism (Definition 3.1, Proposition 3.1, Lemma 3.2).
4. The contrast between the non-injectivity of σ_R and the injectivity of Φ_R has been collected in §3.4.

4.2 Position

The radial projection σ_R defined in this note is identical to the mapping classically known in topology as the **radial projection** (see e.g. [Hatcher, Ch. 0]). The contribution of this note is not the discovery of a new geometric theorem; it is limited to

- explicitly naming and fixing the notation for this foundational mapping as the "radial projection σ_R", and
- making explicit its relation to the operations appearing in the author's central projection framework [1], [2].

This note is released as a **Zenodo preprint / technical note** to consolidate the foundations of the central projection series; submission to a pure-mathematics journal is not intended.

4.3 What This Note Does Not Claim

- That the radial projection σ_R was "undiscovered in prior work" (it is classically known as the radial projection).
- Any deep mathematical result about σ_R (general correspondences with submanifold theory, polar actions, isoparametric functions, conformality, and so on).
- That the rich geometric structure of the central projection Φ_R (induced metric, curvature tensors, etc.) is automatically inherited by σ_R via the mere domain extension (in fact, since injectivity is lost, the pullback is well-defined only when restricted to Π_R in the relevant non-degenerate sense).
- Any physical application, cosmological interpretation, or quantum-theoretic implication.

These are addressed in separate notes or separate papers.

References

[1] Kihara, N. (2026). *Geometric Formulation of Four-Dimensional Space via Central Projection.* Zenodo. Concept DOI: 10.5281/zenodo.19427780.

[2] Kihara, N. (2026). *Composition of Central Projections and a Closed Form for the Composite Curvature Radius: An Algebraic Formulation of Higher-Dimensional Reduction via One Central Projection and Commuting Sections on the Sphere.* Zenodo. Concept DOI: 10.5281/zenodo.20060728.

[3] A. Hatcher (2002). *Algebraic Topology.* Cambridge University Press. (Chapter 0: the deformation retract $r(x) = x/\|x\|$.)

Author: Noriaki Kihara Affiliation: WF System Co., Ltd. ORCID: 0009-0004-6753-4020

Revision History

- **v1 (2026-05-30, first draft)**: initial version.
- **v2 (2026-05-30, revision)**: integrated the comments from a 4-AI peer review (Claude.ai, ChatGPT, Gemini, Grok):
  - Naming: keep "radial projection σ_R" while stating in §1 that this is "the same as the topological radial projection".
  - Position: explicitly framed as a technical note / Zenodo preprint.
  - Image of the central projection explicitly identified as the open upper hemisphere $S^n_+(R)$ (Proposition 3.1).
  - "Angle preservation" was tightened to genuine preservation of angles between vectors (Proposition 2.5, inner-product formula) and the scale-invariance statement was separated out (Proposition 2.6).
  - New propositions added: idempotency (Proposition 2.2), quotient space (Proposition 2.3), kernel of the differential (Proposition 2.4).
  - Claims about being a "higher-level concept" or about "the absence of a standard text" were weakened.
  - The previous Corollary 3.1(ii) was downgraded to Remark 3.1 and made self-contained.
  - Added the external reference Hatcher (reference [3]).
  - The symbol σ_S was renamed κ_S for the section operation of [2].
  - "Phase" replaced by "directional component" throughout.
  - The case $t < 0$ is now treated in a footnote.
  - The orientation mismatch between the central projection Φ_R and the classical gnomonic projection is now explicitly noted in §1.2 and §3.1.
  - Well-definedness is now stated immediately after the definition.
- **v3 (2026-05-30, minor revision)**: reflected the second-round comments from ChatGPT on v2:
  - The proof of the kernel in Proposition 2.4 was rewritten correctly (now in the "equivalence between $D\sigma_R|_x(v) = 0$ and $v = \lambda x$" form).
  - In Remark 3.1 (§3.5), "the pullback is not well-defined" was corrected to "the pullback is well-defined but degenerate" (the pullback of a tensor is always well-defined, independently of injectivity; the issue is degeneracy).
  - The caption of Fig. 3 was corrected so that it refers only to Proposition 2.6 (Proposition 2.5 removed).
  - The caption of Fig. 2 was supplemented to note that the green pair C, C' is not in the image of the central projection.
  - In §1.1, the phrasing "no standalone document exists for it as a primary named operation" was weakened to "useful as a reference base within the series".
  - The caption of Fig. 2 was supplemented with "for $n = 1$ this is $y = R$, and in general it is $x_{n+1} = R$ as in (3.1)" (per Gemini's second-round comment).
- **v3.1 (2026-05-30, minor revision)**: reflected the minor comments from Claude.ai's second-round review of v2:

  - Added to Proposition 2.4 the image of the differential $\mathrm{Im}(D\sigma_R|_x) = x^\perp = T_{\sigma_R(x)}S^n(R)$ (equation (2.4)). This makes the submersion structure more explicit and records that the tangent space of the sphere at $\sigma_R(x)$ equals the orthogonal complement of the radial direction.
- **v3.2 (2026-06-02, copy-editing)**: reflected copy-editing and polishing comments from peer review:
  - Resolved a duplicate equation number: the angle-preservation identity in Proposition 2.5 was renumbered from (2.4) to **(2.5)** (it had collided with the image equation (2.4) of Proposition 2.4).
  - The cross-reference in §1.2 was already correct ("Lemma 3.2"); verified consistent with §3.3 and §4.1.
  - Added to the proof of Proposition 2.4 the **argument for the image** $\mathrm{Im} = x^\perp$ (via $x \cdot D\sigma_R|_x(v) = 0$ and the rank–nullity theorem); previously only the kernel was justified.
  - Renamed the heading of Proposition 2.4 from "Kernel of the differential" to "**Kernel and image of the differential**" (consistent with its content); §4.1 item 2 updated likewise.
  - Tightened the Φ_R "radial direction" cell of the §3.4 table from "no radial direction in the domain" to "**the kernel direction is not contained in the domain's tangent space (transversal)**".
  - Clarified the opening of the Proposition 2.4 proof: replaced the ambiguous $\frac{d}{dx}$ notation with a **directional derivative along** v.